

1. (a) $\frac{\partial g}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{g(h,0) - g(0,0)}{h} = \lim_{h \rightarrow 0} 0 = 0$
 $\frac{\partial g}{\partial y}(0,0) = \lim_{h \rightarrow 0} \frac{g(0,h) - g(0,0)}{h} = \lim_{h \rightarrow 0} 0 = 0$
 (b) As (x, y) tends to $(0,0)$ along the x -axis, $g(x, y) = g(x, 0) = 0$ tends to 0 ;
 As (x, y) tends to $(0,0)$ along the line $y = x$, $g(x, y) = g(x, x) = \frac{1}{2}$ tends to $\frac{1}{2}$.
 Therefore, $\lim_{(x,y) \rightarrow (0,0)} g(x, y)$ does not exist.
2. (a) As (x, y) tends to $(0,0)$ along the x -axis, $\lim_{x \rightarrow 0} f(x, 0) = \lim_{x \rightarrow 0} 0 = 0$
 (b) As (x, y) tends to $(0,0)$ along the y -axis, $\lim_{y \rightarrow 0} f(0, y) = \lim_{y \rightarrow 0} 0 = 0$
 (c) As (x, y) tends to $(0,0)$ along the $y = mx$, $\lim_{x \rightarrow 0} \frac{2x^2mx}{x^4 + m^2x^2} = \lim_{x \rightarrow 0} \frac{2mx}{m^2 + x^2} = 0$
 (d) As (x, y) tends to $(0,0)$ along the $y = ax^2$, $\lim_{x \rightarrow 0} \frac{2x^2ax^2}{x^4 + x^2x^4} = \frac{2a}{1 + a^2}$;
 Therefore, $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.
3. (a) $\nabla f = \frac{2x}{x^2 + y^2} \mathbf{i} + \frac{2y}{x^2 + y^2} \mathbf{j} \Rightarrow$ at $(2, 1)$, $\nabla f = \frac{4}{5} \mathbf{i} + \frac{2}{5} \mathbf{j}$
 (b) $\nabla f = \frac{1}{(x - y)^2} (-2y \mathbf{i} + 2x \mathbf{j}) \Rightarrow$ at $(3, 1)$, $\nabla f = -\frac{1}{2} \mathbf{i} + \frac{3}{2} \mathbf{j}$
4. (a) $\nabla f = (e^y - ye^x) \mathbf{i} + (xe^y - e^x) \mathbf{j} \Rightarrow \nabla f(1, 0) = \mathbf{i} + (1 - e) \mathbf{j}$ and $\mathbf{u} = \frac{1}{5}(3\mathbf{i} + 4\mathbf{j})$
 Therefore, $f'_u(1, 0) = \nabla f(1, 0) \cdot \mathbf{u} = \frac{1}{5}(7 - 4e)$
 (b) $\nabla f = \frac{1}{(x - y)^2} (-2y \mathbf{i} + 2x \mathbf{j}) \Rightarrow \nabla f(1, 0) = 2\mathbf{j}$ and $\mathbf{u} = \frac{1}{2}(\mathbf{i} - \sqrt{3}\mathbf{j})$
 Therefore, $f'_u(1, 0) = \nabla f(1, 0) \cdot \mathbf{u} = -\sqrt{3}$
5. (a) $\nabla f = 2y^2 e^{2x} \mathbf{i} + 2ye^{2x} \mathbf{j} \Rightarrow \nabla f(0, 1) = 2\mathbf{i} + 2\mathbf{j}$, $\|\nabla f\| = 2\sqrt{2}$ and $\frac{\nabla f}{\|\nabla f\|} = \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j})$
 f increases most rapidly in the direction $\mathbf{u} = \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j})$; the rate of change is $2\sqrt{2}$.
 f decreases most rapidly in the direction $\mathbf{v} = -\frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j})$; the rate of change is $-2\sqrt{2}$.
 (b) $\nabla f = [1 + \cos(x + 2y)] \mathbf{i} + 2 \cos(x + 2y) \mathbf{j}$
 $\Rightarrow \nabla f(0, 0) = 2\mathbf{i} + 2\mathbf{j}$, $\|\nabla f\| = 2\sqrt{2}$ and $\frac{\nabla f}{\|\nabla f\|} = \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j})$
 f increases most rapidly in the direction $\mathbf{u} = \frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j})$; the rate of change is $2\sqrt{2}$.
 f decreases most rapidly in the direction $\mathbf{v} = -\frac{1}{\sqrt{2}}(\mathbf{i} + \mathbf{j})$; the rate of change is $-2\sqrt{2}$.